Exercise 18 — Copy a Database Table

SAP S/4HANA Cloud ABAP / RAP — Practical Documentation

Student Group: 09

Exercise Objective

The purpose of this exercise is to prepare the database foundation for an OData UI service that will later be used to change flight prices. The exercise creates a dedicated RAP package, copies the supplied flight database table, and uses a copied ABAP console class to populate the new table with the template flight data.

Development Objects Used

Object Type	Object	Purpose
Main Package	ZS4D400_9309	Student development package
Subpackage	ZS4D400_9309_RAP	Objects for OData UI Service
Template Database Table	/LRN/S4D400_APT	Source table
Copied Database Table	Z09FLIGHT	Student's flight table
Template Global Class	/LRN/CL_S4D400_APT_COPY	Source data-copy program
Copied Global Class	ZCL_09_COPY	Program used to fill Z09FLIGHT

Task 1 — Create Subpackage

A separate subpackage was created under the existing student package so that the development objects for the OData UI service are kept together.

Steps completed:

1. Opened package ZS4D400_9309 in Project Explorer.
2. Created a new ABAP package using New → ABAP Package.
3. Entered ZS4D400_9309_RAP as the package name.
4. Entered Objects for OData UI Service as the description.
5. Set ZS4D400_9309 as the superpackage.
6. Added the package to the favorite packages.
7. Confirmed the software component ZSTUDENTS and completed the transport request.

Result: The subpackage ZS4D400_9309_RAP was successfully created and activated.

Task 2 — Copy Database Table

The supplied database table /LRN/S4D400_APT was used as the template. A duplicate was created in the new RAP subpackage and renamed Z09FLIGHT.

Steps completed:

8. Opened /LRN/S4D400_APT from the Course S4D400 exercise package.
9. Used the Duplicate option from the database table context menu.

10. Selected ZS4D400_9309_RAP as the target package.
11. Entered Z09FLIGHT as the new database table name.
12. Confirmed the transport request.
13. Activated the copied database table with Ctrl + F3.

The resulting table contains the flight-related fields required by the exercise, including:

- CLIENT
- CARRIER_ID
- CONNECTION_ID
- FLIGHT_DATE
- PRICE
- CURRENCY_CODE
- PLANE_TYPE_ID
- LOCAL_CREATED_BY / LOCAL_CREATED_AT
- LOCAL_LAST_CHANGED_BY / LOCAL_LAST_CHANGED_AT
- LAST_CHANGED_AT

Important point: Z09FLIGHT is an exact copy of the supplied template table structure. This is required because the provided copy class expects the target table to have the same type and structure as the template.

Task 3 — Fill Database Table

The supplied class /LRN/CL_S4D400_APT_COPY was copied and adapted so that it writes the template flight data into the newly created Z09FLIGHT table.

Steps completed:

14. Opened /LRN/CL_S4D400_APT_COPY from the Source Code Library.
15. Created the student copy ZCL_09_COPY in package ZS4D400_9309.
16. Opened the copied class and verified that it implements IF_OO_ADT_CLASSRUN.
17. Changed the target table name constant to Z09FLIGHT.
18. Activated the class with Ctrl + F3.
19. Executed the class as a console application with F9.
20. Verified the ABAP Console output after execution.

Important Code Change

```
CONSTANTS table_name TYPE tabname VALUE 'Z09FLIGHT'.
```

The constant is the key customization in the copied program. The original template program is designed to copy flight data into a group-specific target table. For this exercise, the target is Z09FLIGHT.

How the Copy Program Works

table_name: Stores the name of the target database table.

NEW lcl_copy_data(table_name): Creates the local helper object responsible for copying the flight data.

copier->copy_data( ): Performs the data-copy operation from the template source to the target table.

out->write(...): Writes the result of the operation to the ABAP Console.

CATCH cx_abap_not_a_table: Handles the situation where the specified target is not a suitable database table.

Execution Result

The class was activated and executed successfully. The ABAP Console was used to verify the execution result, and the target table Z09FLIGHT was populated with the template flight data.

Expected successful console message:

```
Z09FLIGHT was filled with data
```

Why the Table Must Be an Exact Copy

The helper class works with the structure of the template flight table. Therefore, the target table must contain compatible fields and data types. If the structure or type differs, the copy program can reject the target table. Keeping Z09FLIGHT as an exact duplicate of /LRN/S4D400_APT ensures compatibility with the supplied program.

Exercise Completion Checklist

Requirement	Status
Create ZS4D400_9309_RAP	Completed
Set description to Objects for OData UI Service	Completed
Set ZS4D400_9309 as superpackage	Completed
Create Z09FLIGHT from /LRN/S4D400_APT	Completed
Activate Z09FLIGHT	Completed
Create ZCL_09_COPY from /LRN/CL_S4D400_APT_COPY	Completed
Set target table to Z09FLIGHT	Completed
Activate ZCL_09_COPY	Completed
Execute ZCL_09_COPY as console application	Completed
Populate Z09FLIGHT with flight data	Completed

Final Solution Summary

Exercise 18 was completed by creating the RAP-specific subpackage ZS4D400_9309_RAP, duplicating /LRN/S4D400_APT as Z09FLIGHT, and duplicating /LRN/CL_S4D400_APT_COPY as ZCL_09_COPY. The copied class was adjusted to target Z09FLIGHT and executed successfully as an ABAP console application. The target database table was populated with the required flight data and is ready for the subsequent OData UI service exercises.

Reference — Final Object Structure

```
ZS4D400_9309

└── ZS4D400_9309_RAP
```

```
    └── Z09FLIGHT


ZS4D400_9309

└── ZCL_09_COPY
```